

An RCD Proof of the $n^2 + 3$ Almost-Rigidity Conjecture

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Abstract

Let (M, ω) be a compact Kähler n -fold, $n \geq 2$, with $\text{Ric}(\omega) \geq \omega$. Chu–Wang–Zhang conjectured that if the $(n^2 + 3)$ -rd eigenvalue of the complex Laplacian is sufficiently close to one, then M is biholomorphic and Gromov–Hausdorff close to $\mathbb{CP}^n$ with its normalized Kähler–Einstein metric. We prove this by a contradiction argument on a possibly collapsed Ricci limit. Strong compactness of scalar Hamiltonian brackets yields a ghost–visible decomposition of the first eigenspace; complementary dimension estimates for the two parts exclude collapse. The noncollapsed limit is then shown to be a smooth homogeneous Kähler manifold and is identified with $\mathbb{CP}^n$ by the exact rigidity theorem of Chu–Wang–Zhang.

Status of the note. This is a proof draft intended for line-by-line verification. Every new estimate used below is stated and proved. Foundational results from Ricci limit and RCD theory are invoked explicitly and cited.

1 Introduction

We use the normalization of [7]. Thus $\omega_{\mathbb{CP}^n}$ denotes the Fubini–Study metric rescaled so that

$$\text{Ric}(\omega_{\mathbb{CP}^n}) = \omega_{\mathbb{CP}^n}.$$

Theorem 1.1 (Conjecture 1.8 of [7]). *For every $n \geq 2$ and every $\varepsilon > 0$, there is $\delta = \delta(n, \varepsilon) > 0$ with the following property. If (M, ω) is a compact Kähler n -fold satisfying*

$$\text{Ric}(\omega) \geq \omega, \quad \lambda_{n^2+3}(-\Delta_\omega) < 1 + \delta,$$

then M is biholomorphic to $\mathbb{CP}^n$ and

$$d_{\text{GH}}((M, \omega), (\mathbb{CP}^n, \omega_{\mathbb{CP}^n})) < \varepsilon.$$

Proof strategy

We argue by contradiction. A counterexample sequence has a normalized measured Gromov–Hausdorff limit X , which is a compact $\text{RCD}(1, 2n)$ space. Spectral convergence leaves on X a space

$$E = \ker(L_X - 2), \quad \dim E \geq n^2 + 3.$$

The essential dimension d of X may initially be smaller than $2n$; write $c = 2n - d$ for the collapse codimension. The first and main part of the proof is to show that $c = 0$.

The complex structures J_i need not converge strongly when the sequence collapses. We therefore retain only the scalar Poisson brackets

$$\langle J_i Df_i, Dg_i \rangle.$$

They converge strongly, after recovery of the functions, to $\{f, g\} = \langle ADf, Dg \rangle$, where A is a skew contraction on the tangent module of X . The Kähler defect identity shows that for every $u \in E$, the field $W_u = ADu$ is divergence-free and weakly Killing. A weak commutator formula shows that W_u preserves every Laplacian eigenspace. Consequently these fields integrate to measure-preserving isometries, and the bracket makes E a finite-dimensional metric Lie algebra.

We split this Lie algebra into the *ghost part* $K_0 = \{u \in E : ADu = 0\}$ and its orthogonal, visible part $\mathfrak{v}$. Ghost gradients commute. Moreover, local splitting maps on the regular set show that their prelimit J_i -rotations can only occupy the c directions lost in the collapse. A matrix-square-root argument, combined with the sharp Poincaré inequality, then gives

$$\dim K_0 \leq 2 \min(c, z), \quad z = \dim \ker A_x.$$

At a principal point where $\text{rank } A_x = 2a$, the visible fields carry an intrinsic presymplectic form. The isotropy algebra acts faithfully and centralizes that form, which yields

$$\dim \mathfrak{v} \leq a^2 + 2a.$$

Since $c + z = 2(n - a)$, these estimates imply, whenever $c > 0$,

$$\dim E \leq a^2 + 2n \leq n^2 + 1,$$

contradicting $\dim E \geq n^2 + 3$. This is the numerical reason that the spectral threshold in the theorem rules out every collapsed limit.

Thus $d = 2n$. Honda's noncollapsed Kähler-limit theorem now upgrades the weak limit A to a parallel orthogonal complex structure and the ghost part disappears. A refinement of the visible dimension bound forces a full $2n$ -dimensional orbit. Rather than assuming that such an orbit is open in a singular space, we use metric differentiability and the area formula to prove that it has positive $\mathcal{H}^{2n}$ -measure. Full-dimensional noncollapse gives $\mathfrak{m} = C\mathcal{H}^{2n}$, so the orbit has positive $\mathfrak{m}$ -measure. The positive-measure orbit theorem for RCD spaces then makes X homogeneous, hence smooth. The exact rigidity theorem of Chu–Wang–Zhang identifies X with $\mathbb{CP}^n$. Finally, Colding volume convergence and the Kähler volume-gap theorem show that the approximating manifolds are themselves biholomorphic to $\mathbb{CP}^n$, while their metrics converge to the normalized Fubini–Study metric. This contradicts the defining counterexample sequence.

The proof is organized in the same order: scalar Hamiltonian compactness in [theorems 3.2](#) and [3.5](#), integration to isometries in [theorem 4.1](#), the ghost and visible bounds in [theorems 5.2](#), [5.4](#) and [6.1](#), exclusion of collapse in Section 7, and identification of the limit and approximants in Sections 8 and 9.

Normalization and external inputs

Let g be the real Riemannian metric associated with ω , and put

$$L = -\Delta_g = 2(-\Delta_\omega). \tag{1.1}$$

All eigenspaces below are real. We normalize reference measures to have mass one. For a compact metric-measure space X , write

$$\mathcal{E}(f, h) = \int_X \langle Df, Dh \rangle \, d\mathbf{m}, \quad L \geq 0,$$

and use the continuous Lipschitz representatives of eigenfunctions.

We use the following established results.

- (i) Measured Gromov–Hausdorff compactness, Mosco and spectral convergence, compactness of functions with bounded Hessian, and weak tensor convergence for sequences with a uniform lower Ricci bound [1, 11, 12].
- (ii) Constancy of the essential dimension of an $\text{RCD}(K, N)$ space [4].
- (iii) Existence and uniqueness of regular Lagrangian flows for Sobolev derivations, together with the metric-velocity identification [2]; and the Lipschitz regularity framework for such flows on RCD spaces [5].
- (iv) The Lie property of the isometry group, the principal orbit theorem, and the structure of homogeneous RCD spaces [10, 9, 14].
- (v) Full-dimensional noncollapse and Honda’s Kähler-limit theorem, especially Theorem 6.19 of [12]; see also [3].
- (vi) Colding’s volume convergence theorem [8], the volume-gap theorem of Zhang with Liu’s appendix [15], and the exact eigenvalue rigidity theorem of Chu–Wang–Zhang [7].

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2 The contradiction limit

Set

$$q = n^2 + 3. \tag{2.1}$$

Suppose that [theorem 1.1](#) fails for some $\varepsilon_0 > 0$. There is a sequence of compact Kähler n -folds (M_i, ω_i) such that

$$\text{Ric}(\omega_i) \geq \omega_i, \quad \lambda_q(-\Delta_{\omega_i}) < 1 + i^{-1}, \tag{2.2}$$

while either $M_i \not\cong \mathbb{CP}^n$, or

$$d_{\text{GH}}((M_i, \omega_i), (\mathbb{CP}^n, \omega_{\mathbb{CP}^n})) \geq \varepsilon_0. \tag{2.3}$$

Write g_i for the real metric, $L_i = -\Delta_{g_i}$, and

$$\mathbf{m}_i = \frac{\text{vol}_{g_i}}{\text{vol}_{g_i}(M_i)}.$$

The Kähler Lichnerowicz estimate and (1.1) give

$$2 \leq \mu_1(M_i) \leq \cdots \leq \mu_q(M_i) < 2 + 2i^{-1}, \tag{2.4}$$

where the μ_j ’s are the eigenvalues of L_i .

Myers' theorem gives a uniform diameter bound. After passing to a subsequence,

$$(M_i, d_i, \mathbf{m}_i) \xrightarrow{\text{pmGH}} (X, d, \mathbf{m}), \quad (2.5)$$

where X is compact, has full-support probability measure, and is an $\text{RCD}(1, 2n)$ space. Spectral convergence and (2.4) yield

$$\lambda_1(L_X) = 2, \quad E := \ker(L_X - 2), \quad \dim E =: m_E \geq q. \quad (2.6)$$

By constancy of dimension there is a single integer $d \leq 2n$ such that the d -regular set has full $\mathbf{m}$ -measure. Define the collapse codimension

$$c = 2n - d. \quad (2.7)$$

3 The limit Hamiltonian structure

3.1 The Kähler defect identity

For a smooth real function u on a Kähler manifold, let

$$Y_u = J\nabla u, \quad S_u = \text{sym } \nabla Y_u = \frac{1}{2} \mathcal{L}_{Y_u} g. \quad (3.1)$$

Lemma 3.1 (Kähler defect identity). *On a compact Kähler manifold,*

$$\int |S_u|^2 + \int (\text{Ric} - g)(Du, Du) = \frac{1}{2} \int (Lu)^2 - \int |Du|^2. \quad (3.2)$$

Consequently, if u_i belongs to a spectral band shrinking to 2, has bounded L^2 -norm, and $\text{Ric}(g_i) \geq g_i$, then

$$\|S_{u_i}\|_{L^2} \longrightarrow 0, \quad \sup_i \|\text{Hess } u_i\|_{L^2} < \infty. \quad (3.3)$$

Proof. Write $H = \text{Hess } u$ as a self-adjoint endomorphism. Since $\nabla J = 0$,

$$S_u = \frac{1}{2}(JH - HJ).$$

The integrated Kähler–Bochner identity, obtained by commuting one $(1, 0)$ and one $(0, 1)$ derivative, is

$$\int |S_u|^2 + \int \text{Ric}(Du, Du) = \frac{1}{2} \int (Lu)^2. \quad (3.4)$$

This is the real form of the calculation used in [7, Lemma 2.1]. Subtracting $\int |Du|^2$ gives (3.2).

If $u_i = \sum_j a_{ij} e_{ij}$, where the e_{ij} 's are orthonormal eigenfunctions and all corresponding eigenvalues μ_{ij} tend uniformly to 2, then the right side of (3.2) is

$$\sum_j a_{ij}^2 \left(\frac{\mu_{ij}^2}{2} - \mu_{ij} \right) \longrightarrow 0.$$

Both terms on the left are nonnegative. This proves the first assertion in (3.3). The ordinary integrated Bochner identity gives

$$\int |\text{Hess } u_i|^2 + \int \text{Ric}(Du_i, Du_i) = \int (L_i u_i)^2,$$

which proves the Hessian bound. $\square$

Eigenfunction gradient estimates under a uniform lower Ricci and diameter bound also give a uniform Lipschitz bound for every fixed finite spectral band. We use this fact repeatedly below.

We fix a countable algebra $\mathcal{A} \subset \text{Test}(X)$ generated by functions $P_t\varphi$, where $t > 0$ is rational and φ ranges over a countable bounded Lipschitz $W^{1,2}$ -core. Using the compatible prelimit heat flows and second-order stability [1, 12], and then taking a diagonal subsequence, every $f \in \mathcal{A}$ has recoveries f_i such that

$$f_i \rightarrow f, \quad Df_i \rightarrow Df, \quad \Delta_i f_i \rightarrow \Delta f$$

strongly in L^2 , and

$$\sup_i (\text{Lip}(f_i) + \|\text{Hess } f_i\|_{L^2}) < \infty.$$

The algebra is chosen dense in the graph norms used below. Eigenfunctions are recovered separately by spectral convergence. All density extensions refer to this fixed core, and all finite families are made simultaneous by the same diagonal choice. The local harmonic recoveries needed on splitting charts are recorded separately in [theorem 5.3](#). We call functions obtained from these cores and their finite spectral closure *admissible*.

3.2 A weak complex structure and strong scalar brackets

Honda's weak tensor compactness gives, after a subsequence, a weak L^2 -limit A of the J_i 's on the d -dimensional tangent module of X . Weak convergence alone is insufficient to pass quadratic identities. The following scalar compactness is the replacement.

Lemma 3.2 (Strong bracket compactness). *There is a measurable tangent-module endomorphism A such that*

$$A^* = -A, \quad \|A\|_{\text{op}} \leq 1. \quad (3.5)$$

For f, g in a countable dense Test algebra, and then by closure for all functions used below, define

$$\{f, g\} := \langle ADf, Dg \rangle. \quad (3.6)$$

If f_i, g_i are Test recoveries with uniformly bounded Lipschitz constants and L^2 -Hessian norms, then

$$\langle J_i Df_i, Dg_i \rangle \longrightarrow \{f, g\} \quad \text{strongly in } L^2. \quad (3.7)$$

Moreover

$$\text{div}(ADf) = 0. \quad (3.8)$$

Proof. Put $q_i = \langle J_i Df_i, Dg_i \rangle$. Since $\nabla J_i = 0$,

$$|Dq_i| \leq \text{Lip}(g_i) |\text{Hess } f_i| + \text{Lip}(f_i) |\text{Hess } g_i|. \quad (3.9)$$

Thus q_i is uniformly bounded in $W^{1,2}$. Scalar Rellich compactness along (2.5) gives strong L^2 -convergence after passing to a subsequence. The gradient recoveries converge strongly in L^2 and are uniformly bounded in L^∞ , hence converge strongly in L^4 . Thus $Df_i \otimes Dg_i \rightarrow Df \otimes Dg$ strongly in L^2 , which identifies the pairing with the weak tensor limit of J_i as $\langle ADf, Dg \rangle$. A diagonal argument on a countable Test algebra therefore proves (3.7). The linear identities $J_i^* = -J_i$ pass weakly, and the convex operator-norm bound passes by lower semicontinuity, giving (3.5).

Finally,

$$\operatorname{div}(J_i Df_i) = \operatorname{tr}(J_i \operatorname{Hess} f_i) = 0$$

because J_i is skew and $\operatorname{Hess} f_i$ is symmetric. Passing the corresponding integration-by-parts identity to the limit proves (3.8). Density extends the construction from the chosen core. $\square$

Remark 3.3. Only the scalar quantities in (3.7) converge strongly. We neither use nor assert strong convergence of J_i , or of $J_i \nabla u_i$, in the collapsed regime. Such a claim is false in general; compare [12, Remarks 6.17–6.18].

3.3 Jacobi and spectral closure

The smooth brackets are Poisson brackets. An integrated form of Jacobi is stable under the scalar convergence just proved.

Lemma 3.4 (Limit Jacobi identity). *For admissible u, f, g, η ,*

$$-\int \{f, g\} \{u, \eta\} = \int \{u, f\} \{g, \eta\} - \int \{u, g\} \{f, \eta\}. \quad (3.10)$$

Equivalently,

$$W_u \{f, g\} = \{W_u f, g\} + \{f, W_u g\} \quad \text{in distributions,} \quad W_u := ADu. \quad (3.11)$$

Proof. The smooth Jacobi identity is

$$\{u, \{f, g\}\} = \{\{u, f\}, g\} + \{f, \{u, g\}\}.$$

Multiply by η and integrate. Each Hamiltonian field is divergence-free, so integration by parts gives exactly (3.10). Every bracket factor occurring there converges strongly in L^2 by (3.9); hence every product converges in L^1 , and the identity passes to the limit. $\square$

We next record the commutator calculation that prevents loss of the eigenspace under the bracket.

Lemma 3.5 (Weak Laplace–Killing commutator). *Let Y be a smooth divergence-free vector field, $p = Yv$, and $Lv = \mu v + s$. Then*

$$\mathcal{E}(p, \phi) - \mu \int p\phi = 2 \int S_Y(Dv, D\phi) - \int sY\phi, \quad S_Y = \operatorname{sym} \nabla Y. \quad (3.12)$$

For every $u \in E$, the limit derivation $W_u = ADu$ preserves every Laplacian eigenspace:

$$W_u(\ker(L_X - \mu)) \subseteq \ker(L_X - \mu). \quad (3.13)$$

In particular,

$$\{E, E\} \subset E. \quad (3.14)$$

Proof. The identity

$$\int \langle D(Yv), D\phi \rangle + \int \langle Dv, D(Y\phi) \rangle = 2 \int S_Y(Dv, D\phi)$$

follows by expanding $\mathcal{L}_Y g$ and using $\operatorname{div} Y = 0$. Since

$$\int \langle Dv, D(Y\phi) \rangle = \int (\mu v + s)Y\phi = -\mu \int (Yv)\phi + \int sY\phi,$$

we obtain (3.12).

Fix $u \in E$ and $v \in \ker(L_X - \mu)$. Choose spectral recoveries u_i, v_i such that u_i lies in a band shrinking to 2, while

$$L_i v_i = \mu v_i + s_i, \quad \|s_i\|_2 \longrightarrow 0.$$

Apply (3.12) with $Y_i = J_i Du_i$. By theorem 3.1, $\|S_{Y_i}\|_2 \rightarrow 0$. The relevant eigenfunction gradients are uniformly bounded, and $\langle J_i Du_i, Dv_i \rangle$ converges strongly by theorem 3.2. Estimate (3.9) also bounds this last sequence in $W^{1,2}$, so its gradients converge weakly after the strong L^2 -limit has been identified. We may therefore pass in the Dirichlet term of (3.12); the two terms on its right converge to zero. This gives

$$\mathcal{E}(\{u, v\}, \phi) = \mu \int \{u, v\} \phi$$

for every Test function ϕ , proving (3.13) and (3.14). $\square$

The same limiting calculation, written in the weak definition of the symmetric covariant derivative, gives

$$\operatorname{div} W_u = 0, \quad D^{\operatorname{sym}} W_u = 0 \tag{3.15}$$

in the Ambrosio–Trevisan sense. More explicitly, put $\Delta = -L$. For Test functions f, g , their weak symmetric deformation functional is

$$\mathcal{D}_b(f, g) = -\frac{1}{2} \int [b(f)\Delta g + b(g)\Delta f - (\operatorname{div} b)\Gamma(f, g)] \, \mathrm{d}\mathbf{m}. \tag{3.16}$$

For the smooth prelimit field $b_i = J_i Du_i$, integration by parts gives

$$\mathcal{D}_{b_i}(f_i, g_i) = \int S_{u_i}(Df_i, Dg_i) \, \mathrm{d}\mathbf{m}_i. \tag{3.17}$$

The quantities $b_i(f_i)$ and $b_i(g_i)$ converge strongly by theorem 3.2, while the Laplacians of Test recoveries converge strongly. Since $\|S_{u_i}\|_2 \rightarrow 0$, (3.16) vanishes for $b = W_u$. This proves (3.15) without asserting membership in a stronger noncollapsed covariant Sobolev class.

4 From spectral derivations to isometric flows

Proposition 4.1 (Integration of a spectral Killing derivation). *Let (Z, d_Z, ν) be a compact $\operatorname{RCD}(K, N)$ space with full-support probability measure, and let $b \in L^\infty(TZ)$ satisfy*

$$\operatorname{div} b = 0, \quad D^{\operatorname{sym}} b = 0, \quad b(\ker(L - \lambda)) \subseteq \ker(L - \lambda) \quad \text{for every } \lambda. \tag{4.1}$$

Then b integrates to a continuous one-parameter group $\Phi_t \in \operatorname{Iso}_\nu(Z)$. If B_λ is the restriction of b to $\ker(L - \lambda)$, its Koopman group satisfies

$$U_t f := f \circ \Phi_t = e^{tB_\lambda} f \quad \text{for } f \in \ker(L - \lambda). \tag{4.2}$$

Proof. The equality $D^{\text{sym}}b = 0$ is understood in the Ambrosio–Trevisan sense: the bilinear deformation functional in (3.16), with ν in place of $\mathfrak{m}$, has norm zero. The structural assumptions on the algebra, the L^p - Γ inequalities, and the distance required by Ambrosio–Trevisan hold on a compact $\text{RCD}(K, N)$ space; see [2, Sections 6 and 9.6]. For existence we apply [2, Theorem 4.3] with $r = \infty$. For uniqueness we apply [2, Theorem 5.4] with $q = 2$ and $r = s = 4$, so that $q^{-1} + r^{-1} + s^{-1} = 1$; the deformation term is zero. Since $\nu(Z) < \infty$, every nonnegative solution in the L^∞ class required by [2, Theorem 8.3] belongs to this uniqueness class. Theorem 8.3 therefore gives unique regular Lagrangian flows for b and $-b$ on every bounded time interval.

For the initial datum $u_0 \equiv 1$, the constant curve $u_t \equiv 1$ is a solution because $\text{div } b = 0$. The density of $(\Phi_t)_\# \nu$ is another solution with the same initial datum; uniqueness gives $(\Phi_t)_\# \nu = \nu$, and similarly for the flow of $-b$. Autonomous uniqueness, applied also after time translation and to the field $-b$, gives, initially modulo null sets,

$$\Phi_{t+s} = \Phi_t \circ \Phi_s, \quad \Phi_{-t} = \Phi_t^{-1}, \quad (\Phi_t)_\# \nu = \nu. \quad (4.3)$$

Thus $U_t f = f \circ \Phi_t$ is positive and multiplicative and is an isometry on every L^p .

Let $e_1, \dots, e_\ell$ be a real orthonormal basis of $H_\lambda := \ker(L - \lambda)$, and write $be_j = \sum_k (B_\lambda)_{jk} e_k$. The RLF chain rule gives in $L^2(Z)^\ell$

$$\frac{d}{dt} U_t e_j = U_t (be_j) = \sum_k (B_\lambda)_{jk} U_t e_k. \quad (4.4)$$

This finite-dimensional constant-coefficient ODE proves (4.2). Integration by parts shows that

$$\langle B_\lambda f, h \rangle_2 = \int (bf)h \, d\nu = - \int f(bh) \, d\nu = -\langle f, B_\lambda h \rangle_2, \quad (4.5)$$

so e^{tB_λ} is orthogonal. Expanding in the discrete spectral resolution,

$$\|U_t f\|_2 = \|f\|_2, \quad \mathcal{E}(U_t f) = \mathcal{E}(f), \quad U_t L = L U_t. \quad (4.6)$$

For bounded $f, h \in W^{1,2}(Z)$,

$$\int h \Gamma(f) \, d\nu = \mathcal{E}(f, hf) - \frac{1}{2} \mathcal{E}(f^2, h). \quad (4.7)$$

Using multiplicativity and (4.6) in (4.7) gives

$$\Gamma(U_t f) = U_t \Gamma(f) \quad \nu\text{-a.e.} \quad (4.8)$$

If f is L_0 -Lipschitz, then $\Gamma(f) \leq L_0^2$; the Sobolev-to-Lipschitz property and (4.8) give an L_0 -Lipschitz representative of $U_t f$.

Choose a countable dense set $\{z_j\} \subset Z$ and put $r_j = d_Z(\cdot, z_j)$. On one conull dense set, the chosen one-Lipschitz representatives of $U_t r_j$ agree with $r_j \circ \Phi_t$. Hence

$$d_Z(\Phi_t x, \Phi_t y) = \sup_j |r_j(\Phi_t x) - r_j(\Phi_t y)| \leq d_Z(x, y). \quad (4.9)$$

Carry out the preceding argument first for $q \in \mathbb{Q}$. Equation (4.9), applied to q and $-q$, and the rational inverse identity extend these maps to mutually inverse isometries of Z . Intersect the

corresponding full-measure sets, also including the rational group identities. The RLF upper-speed estimate [2, Lemmas 7.4 and 9.2] gives on this common set

$$d_Z(\Phi_q x, \Phi_r x) \leq \|b\|_\infty |q - r| \quad (q, r \in \mathbb{Q}).$$

The rational-time maps extend to isometries, so the same inequality holds for every $x \in Z$. Thus $q \mapsto \Phi_q$ is Cauchy in the uniform metric as $q \rightarrow t$; define Φ_t to be its uniform limit. The rational group and inverse identities pass to the limit, and

$$\sup_{x \in Z} d_Z(\Phi_t x, \Phi_s x) \leq \|b\|_\infty |t - s|.$$

For each fixed t , this extension agrees ν -almost everywhere with the original RLF time- t map, by continuity of almost every RLF trajectory and rational approximation. Hence it preserves ν and still satisfies (4.2). The last display and the isometry property give joint continuity in (t, x) . $\square$

Applying theorem 4.1 to (3.15) and (3.13) produces a measure-preserving isometric flow Φ_t^u for every $u \in E$.

These one-parameter groups fit together into a Lie-group action. On the finite spectral core, theorem 3.4 gives

$$[W_u, W_v] = W_{\{u, v\}}. \quad (4.10)$$

For each Laplace eigenspace H_λ , the operators $B_\lambda(u) := W_u|_{H_\lambda}$ therefore form a finite-dimensional skew representation of E . Let G_{sc} be the simply connected Lie group integrating E . These representations integrate uniquely to orthogonal representations $\rho_\lambda : G_{\text{sc}} \rightarrow O(H_\lambda)$, whose Hilbert direct sum is a unitary representation on $L^2(X)$.

Every element of G_{sc} is a finite product of exponentials. By (4.4), its unitary operator is the corresponding finite product of the actual Koopman operators from theorem 4.1, hence is pull-back by a measure-preserving isometry of X . If two exponential products represent the same group element, their pull-backs agree on every eigenspace, hence on the dense spectral core and on $L^2(X)$; full support then makes the two continuous isometries equal pointwise. Thus the isometry is independent of the product. Local product-of-exponentials coordinates and continuity of the one-parameter flows make the resulting action continuous.

Let T_g denote the isometry whose pull-back is $\rho(g)$. Since pull-back is contravariant,

$$T_{gh} = T_h \circ T_g.$$

Thus $g \mapsto T_g$ is an antihomomorphism. To use the standard Lie algebra convention for $\text{Iso}_{\mathfrak{m}}(X)$, set

$$j(u) = \left. \frac{d}{dt} \right|_{t=0} T_{\exp(-tu)} \in \text{Lie Iso}_{\mathfrak{m}}(X).$$

The differential of an antihomomorphism is a Lie-algebra antihomomorphism, so the extra minus sign makes j a homomorphism. Moreover,

$$\ker j = \{u \in E : W_u = 0\}, \quad \exp_{\text{Iso}_{\mathfrak{m}}(X)}(tj(u)) = \Phi_{-t}^u.$$

After the ghost-visible splitting is introduced in (5.2), we identify the faithful algebra $\mathfrak{v}$ with $j(\mathfrak{v}) \subset \text{Lie Iso}_{\mathfrak{m}}(X)$. All brackets and adjoint actions below use this identification. We retain the convenient notation $e_x(u) = W_u(x)$; the sign is immaterial for speeds and kernels, and the transported identification gives (6.6) with its displayed sign. The image is an analytic subgroup, not necessarily closed, and its compact closure will be used whenever closedness is required.

Lemma 4.2 (A common speed set). *Let $\mathfrak{g}$ be finite dimensional and let $\xi \mapsto b_\xi \in L^\infty(TZ)$ be linear, with*

$$\|b_\xi\|_\infty \leq C\|\xi\|. \quad (4.11)$$

Assume the flows supplied by [theorem 4.1](#) fit together into a continuous left or right action of a connected Lie group. There is one conull Borel set Z_ such that, for all $x \in Z_*$ and all $\xi \in \mathfrak{g}$,*

$$N_x(\xi) := \lim_{t \rightarrow 0} \frac{d_Z(\Phi_t^\xi x, x)}{|t|} = |b_\xi(x)|. \quad (4.12)$$

On this set,

$$b_\xi(x) = 0 \iff \Phi_t^\xi x = x \text{ for every } t \in \mathbb{R}. \quad (4.13)$$

Proof. Fix $\xi \in \mathfrak{g}$ and put

$$a_x^\xi(h) = d_Z(\Phi_h^\xi x, x).$$

For almost every x , the RLF curve $\gamma_x(s) = \Phi_s^\xi x$ is absolutely continuous. The metric-velocity identification [2, Lemmas 7.4 and 9.2] gives

$$|\dot{\gamma}_x|(s) = |b_\xi|(\Phi_s^\xi x) \text{ for a.e. } (x, s). \quad (4.14)$$

The group law and isometry property also give

$$d_Z(\gamma_x(s+h), \gamma_x(s)) = a_x^\xi(h), \quad (4.15)$$

so the difference quotient is independent of s . The metric derivative exists for almost every s ; hence

$$N_x(\xi) := \lim_{h \rightarrow 0} \frac{a_x^\xi(h)}{|h|}$$

exists for almost every x . Whenever it exists, stationarity also gives $N_{\Phi_{s_0}^\xi x}(\xi) = N_x(\xi)$. By Fubini, choose s_0 such that for almost every x the metric-velocity identity holds at s_0 . Then $N_x(\xi) = |b_\xi|(\Phi_{s_0}^\xi x)$ for almost every x . Stationarity and $(\Phi_{s_0}^\xi)_\# \nu = \nu$, followed by the change of variables $y = \Phi_{s_0}^\xi x$, give

$$N_y(\xi) = |b_\xi(y)| \text{ for } \nu\text{-a.e. } y.$$

Here we choose this precise representative of $|b_\xi|$.

We now make the exceptional set independent of ξ . Choose a basis of $\mathfrak{g}$, simultaneous fiber representatives of the corresponding fields, and a countable dense rational subspace $\mathfrak{g}_\mathbb{Q}$. Intersecting the preceding full-measure sets for $\xi \in \mathfrak{g}_\mathbb{Q}$ gives a conull Borel set Z_* . The RLF upper-speed bound and (4.11) give, first almost everywhere and then everywhere by continuity,

$$d_Z(\Phi_t^\xi y, y) \leq C|t| \|\xi\|. \quad (4.16)$$

In local product-of-exponentials coordinates this makes every orbit map locally Lipschitz in the group variable, uniformly in the base point. Thus, for ξ, η in a fixed bounded set and small t ,

$$\begin{aligned} \left| \frac{d_Z(\Phi_t^\xi x, x)}{|t|} - \frac{d_Z(\Phi_t^\eta x, x)}{|t|} \right| &\leq \frac{d_Z(\Phi_t^\xi x, \Phi_t^\eta x)}{|t|} \\ &\leq C_R \|\xi - \eta\|, \end{aligned} \quad (4.17)$$

because $d_G(\exp(t\xi), \exp(t\eta)) \leq C_R|t|\|\xi - \eta\|$. The difference quotients are therefore equi-Lipschitz in ξ . Rational approximation proves simultaneously that the limit exists and equals $|b_\xi(x)|$ for every $x \in Z_*$ and every $\xi \in \mathfrak{g}$.

If the flow fixes x , then $N_x(\xi) = 0$. Conversely, put $a(t) = d_Z(\Phi_t^\xi x, x)$. If $N_x(\xi) = 0$, then for every T and positive integer m , the group law and isometry property give

$$a(T) \leq m a(T/m) = |T| \frac{a(T/m)}{|T|/m} \longrightarrow 0. \quad (4.18)$$

Hence $\Phi_T^\xi x = x$ for every T . $\square$

5 The ghost eigenspace

The bracket (3.6), restricted to E , makes E a finite-dimensional metric Lie algebra. Indeed, Jacobi follows from theorem 3.4, and (3.8) gives

$$\langle \{u, v\}, w \rangle_{L^2} = -\langle v, \{u, w\} \rangle_{L^2}. \quad (5.1)$$

Define

$$K_0 = \{u \in E : ADu = 0\}, \quad \mathfrak{v} = K_0^\perp \subset E. \quad (5.2)$$

Since $\{u, v\} = 0$ whenever $u \in K_0$, the space K_0 is central. If $u, v \in \mathfrak{v}$ and $k \in K_0$, then

$$\langle \{u, v\}, k \rangle_2 = -\langle v, \{u, k\} \rangle_2 = 0.$$

Thus $\mathfrak{v} = K_0^\perp$ is a Lie ideal. With the restricted L^2 -inner product it is a compact metric Lie algebra, and its realization by the fields W_u is faithful.

5.1 Ghost gradients commute

Lemma 5.1. *If $u, v \in K_0$, then*

$$[Du, Dv] = 0 \quad \text{m-a.e.} \quad (5.3)$$

Proof. Choose spectral recoveries u_i, v_i in the band $[2, 2 + o(1)]$, and put

$$p_i = \langle J_i Du_i, Dv_i \rangle.$$

Since $ADu = ADv = 0$, theorem 3.2 gives $p_i \rightarrow 0$ strongly in L^2 . Apply (3.12) with $Y_i = J_i Du_i$, $\phi = p_i$, and $L_i v_i = 2v_i + s_i$, where $\|s_i\|_2 \rightarrow 0$. Uniform gradient bounds and $\|S_{u_i}\|_2 \rightarrow 0$ give

$$\|Dp_i\|_2^2 \leq 2\|p_i\|_2^2 + o(1)\|Dp_i\|_2,$$

and hence

$$\|Dp_i\|_2 \longrightarrow 0. \quad (5.4)$$

Let H_{u_i}, H_{v_i} denote the Hessians as self-adjoint endomorphisms. Direct differentiation of p_i gives the exact identity

$$\begin{aligned} Dp_i - J_i[Du_i, Dv_i] &= (H_{v_i} J_i - J_i H_{v_i}) Du_i \\ &\quad - (H_{u_i} J_i - J_i H_{u_i}) Dv_i. \end{aligned} \quad (5.5)$$

The two commutator terms have L^2 -norm tending to zero by (3.1) and (3.3). Together with (5.4) and the orthogonality of J_i , this yields

$$\|[Du_i, Dv_i]\|_2 \longrightarrow 0.$$

Weak Hessian convergence and strong gradient convergence identify the weak limit with $[Du, Dv]$ by the second-order stability of spectral recoveries [1, 12], proving (5.3). $\square$

5.2 A matrix-square-root estimate

The following lemma is the quantitative core of the ghost estimate.

Lemma 5.2 (Commuting-gradient matrix lemma). *Let $u_1, \dots, u_M \in \ker(L_X - 2)$ be L^2 -orthonormal. Assume*

$$[Du_i, Du_j] = 0 \quad \text{for all } i, j, \quad (5.6)$$

and

$$\text{rank span}\{Du_1(x), \dots, Du_M(x)\} \leq r \quad \text{for a.e. } x. \quad (5.7)$$

Then

$$M \leq 2r. \quad (5.8)$$

Proof. Define

$$T_x : T_x X \longrightarrow \mathbb{R}^M, \quad T_x(\xi) = (du_1(\xi), \dots, du_M(\xi)),$$

and

$$G_x = T_x T_x^*, \quad F_x = G_x^{1/2}. \quad (5.9)$$

Since $\int \langle Du_i, Du_j \rangle \, d\mathbf{m} = 2\delta_{ij}$,

$$\int_X G_x \, d\mathbf{m}(x) = 2I_M, \quad \int_X |F_x|_{\text{HS}}^2 \, d\mathbf{m}(x) = 2M. \quad (5.10)$$

The entries $G_{ij} = \Gamma(u_i, u_j)$ belong to $W^{1,2}(X)$. For $\epsilon > 0$, the map

$$\Phi_\epsilon(B) = (B + \epsilon I)^{1/2}$$

is smooth on the open set $\{B \in \text{Sym}_M : B > -\epsilon I\}$, which contains the essential range of G . Hence the matrix-valued Sobolev chain rule applies to $F_\epsilon = \Phi_\epsilon(G)$. We estimate its differential uniformly in ϵ .

We first prove

$$\int_X |DF|_{\text{HS}}^2 \, d\mathbf{m} \leq \sum_{j=1}^M \int_X |\text{Hess } u_j|^2 \, d\mathbf{m} \leq 2M. \quad (5.11)$$

For the last inequality, choose simultaneous orthonormal spectral recoveries $u_{j,i} \rightarrow u_j$, with eigenvalues $\mu_{j,i} \rightarrow 2$. The smooth Bochner identity and $\text{Ric}(g_i) \geq g_i$ give

$$\int_{M_i} |\text{Hess } u_{j,i}|^2 \, d\mathbf{m}_i \leq \mu_{j,i}^2 - \mu_{j,i}.$$

Weak Hessian convergence and lower semicontinuity therefore yield

$$\int_X |\text{Hess } u_j|^2 \, d\mathbf{m} \leq \liminf_{i \rightarrow \infty} (\mu_{j,i}^2 - \mu_{j,i}) = 2.$$

For the first inequality, use measurable orthonormal module frames. At almost every point choose singular-vector frames for T , and write $s = \text{rank } T_x$, so that

$$T_{j\alpha} = \sigma_j \delta_{j\alpha}, \quad \sigma_j > 0 \quad (1 \leq j \leq s),$$

while $Du_j(x) = 0$ for $j > s$. We evaluate the already defined differential in these pointwise frames; no differentiation of the chosen frames is involved.

If $i \leq s < j$, the commuting-gradient identity gives

$$H_{u_j} Du_i = H_{u_i} Du_j = 0.$$

Consequently $D_\beta G_{ij} = 0$; the same conclusion is immediate when $i, j > s$. Thus DG , and hence DF_ϵ , has no entries joining the positive spectral subspace of G to its kernel.

It remains to consider $1 \leq i, j \leq s$. For a tangent direction e_β , put

$$C_{j\alpha}^\beta = (\nabla_{e_\beta} T)_{j\alpha} = \text{Hess } u_j(e_\beta, e_\alpha).$$

Then

$$D_\beta G_{ij} = C_{ij}^\beta \sigma_j + \sigma_i C_{ji}^\beta.$$

The commuting-gradient identity gives

$$C_{ij}^\beta \sigma_j = \sigma_i C_{ji}^\beta. \quad (5.12)$$

Hence $D_\beta G_{ij} = 2C_{ij}^\beta \sigma_j$. Differentiating $F_\epsilon^2 = G + \epsilon I$ gives a Sylvester equation, and therefore

$$(D_\beta F_\epsilon)_{ij} = \frac{2C_{ij}^\beta \sigma_j}{\sqrt{\sigma_i^2 + \epsilon} + \sqrt{\sigma_j^2 + \epsilon}}, \quad 1 \leq i, j \leq s, \quad (5.13)$$

while every entry involving the kernel of G vanishes. The diagonal terms are bounded by $(C_{ii}^\beta)^2$. Pairing the (i, j) and (j, i) terms and using (5.12), the required estimate reduces, after setting $t = \sigma_i/\sigma_j$, to

$$\frac{8t^2}{(1+t)^2} \leq 1 + t^2. \quad (5.14)$$

There is no hidden estimate here:

$$(1+t^2)(1+t)^2 - 8t^2 = (t-1)^2(t^2+4t+1) \geq 0. \quad (5.15)$$

The denominator in (5.13) only increases when $\epsilon > 0$. Summing over i, j, β gives

$$|DF_\epsilon|_{\text{HS}}^2 \leq |\nabla T|^2 = \sum_j |\text{Hess } u_j|^2.$$

The matrices F_ϵ converge pointwise and strongly in L^2 to F , while the displayed estimate gives a uniform $W^{1,2}$ bound. Weak lower semicontinuity as $\epsilon \downarrow 0$ proves that $F \in W^{1,2}$ and yields the first inequality in (5.11).

Let

$$\bar{F} = \int_X F_x \, d\mathbf{m}(x).$$

The Poincaré inequality with sharp constant $\lambda_1(X) = 2$, applied entrywise, and (5.11) give

$$2 \int |F - \bar{F}|_{\text{HS}}^2 \leq \int |DF|_{\text{HS}}^2 \leq 2M.$$

Using (5.10),

$$|\bar{F}|_{\text{HS}}^2 = 2M - \int |F - \bar{F}|_{\text{HS}}^2 \geq M. \quad (5.16)$$

Let P_x be the support projection of F_x . Define operators

$$\begin{aligned}\mathcal{A} : \mathbb{R}^M &\rightarrow L^2(X; \mathbb{R}^M), & (\mathcal{A}v)(x) &= P_x v, \\ \mathcal{C} : \mathbb{R}^M &\rightarrow L^2(X; \mathbb{R}^M), & (\mathcal{C}v)(x) &= F_x v.\end{aligned}$$

Since $P_x F_x = F_x P_x = F_x$,

$$\mathcal{C}^* \mathcal{A} = \mathcal{A}^* \mathcal{C} = \bar{F}, \quad \mathcal{C}^* \mathcal{C} = 2I_M, \quad \mathcal{A}^* \mathcal{A} = \int_X P_x \, \mathrm{d}\mathbf{m}. \quad (5.17)$$

Therefore

$$\bar{F}^2 = \mathcal{A}^* \mathcal{C} \mathcal{C}^* \mathcal{A} \leq \|\mathcal{C}\|^2 \mathcal{A}^* \mathcal{A} = 2 \int_X P_x \, \mathrm{d}\mathbf{m}.$$

Taking traces and using (5.7),

$$|\bar{F}|_{\mathrm{HS}}^2 \leq 2 \int_X \mathrm{rank} P_x \, \mathrm{d}\mathbf{m} \leq 2r. \quad (5.18)$$

Combining (5.16) and (5.18) proves (5.8). $\square$

5.3 The vertical-rank lemma

Lemma 5.3 (Local harmonic splitting recoveries). *Let $\mathcal{R}_d(X)$ be the d -regular set of X . After one further diagonal subsequence, there are countably many points $x_\ell \in X$, radii $r_\ell > 0$, Borel sets*

$$V_\ell \subset B_{r_\ell}(x_\ell), \quad \mathbf{m}\left(\mathcal{R}_d(X) \setminus \bigcup_{\ell=1}^{\infty} V_\ell\right) = 0,$$

and harmonic maps

$$h_\ell = (h_\ell^1, \dots, h_\ell^d) : B_{2r_\ell}(x_\ell) \longrightarrow \mathbb{R}^d$$

with the following properties. For suitable $x_{\ell,i} \in M_i$ converging to x_ℓ , there are smooth harmonic maps

$$h_{\ell,i} = (h_{\ell,i}^1, \dots, h_{\ell,i}^d) : B_{2r_\ell}(x_{\ell,i}) \longrightarrow \mathbb{R}^d$$

such that, for every fixed ℓ ,

- (a) $h_{\ell,i} \rightarrow h_\ell$ locally uniformly and strongly in $W^{1,2}$;
- (b) on the concentric ball of radius $3r_\ell/2$,

$$\sup_i \max_{\alpha} \left(\|Dh_{\ell,i}^{\alpha}\|_{L^{\infty}(B_{3r_\ell/2}(x_{\ell,i}))} + \|\mathrm{Hess} h_{\ell,i}^{\alpha}\|_{L^2(B_{3r_\ell/2}(x_{\ell,i}), \mathbf{m}_i)} \right) < \infty;$$

- (c) if

$$G_{\ell,i}^{\alpha\beta} = \langle Dh_{\ell,i}^{\alpha}, Dh_{\ell,i}^{\beta} \rangle, \quad G_{\ell}^{\alpha\beta} = \langle Dh_{\ell}^{\alpha}, Dh_{\ell}^{\beta} \rangle,$$

then $G_{\ell,i} \rightarrow G_{\ell}$ strongly in local L^p for every finite p , and $G_{\ell}(x)$ is positive definite for $\mathbf{m}$ -almost every $x \in V_{\ell}$.

Proof. This is the standard local harmonic-splitting construction on the regular set of a Ricci limit. At almost every d -regular point, choose a scale at which the ball is arbitrarily close to a d -dimensional Euclidean ball, construct a harmonic splitting map on a slightly larger ball, and take its smooth harmonic recoveries on M_i . Interior gradient bounds and the cutoff Bochner estimate give (b). Strong $W^{1,2}$ -convergence of the harmonic functions, together with the uniform

Lipschitz bound, gives the asserted L^p -convergence of their Gram matrices. Choose the splitting accuracy along a sequence $\varepsilon_k \downarrow 0$. On each selected ball retain the subset on which the Gram matrix of the normalized splitting map has smallest eigenvalue at least $1/2$; the complement has relative measure $O(\varepsilon_k)$. A countable Vitali selection, followed by diagonal exhaustion as $k \rightarrow \infty$, produces Borel sets V_ℓ covering $\mathcal{R}_d(X)$ up to an $\mathfrak{m}$ -null set. These are the standard harmonic-splitting, rectifiability, and second-order compactness results of [6, 11, 12, 1]. $\square$

Lemma 5.4 (Ghost gradients are vertical). *Let*

$$r_g = \operatorname{ess\,sup}_{x \in X} \operatorname{rank} \operatorname{span}\{Du(x) : u \in K_0\}. \quad (5.19)$$

Then

$$r_g \leq c = 2n - d. \quad (5.20)$$

At every point where A_x and the ghost gradients have Lebesgue values,

$$\operatorname{rank} \operatorname{span}\{Du(x) : u \in K_0\} \leq \dim \ker A_x. \quad (5.21)$$

Proof. The second assertion follows immediately from $ADu = 0$: at every common Lebesgue point, the span of the ghost gradients lies in $\ker A_x$. We prove (5.20).

Fix $u^1, \dots, u^{c+1} \in K_0$. It is enough to show that their gradients are linearly dependent almost everywhere. Fix a chart from theorem 5.3, suppress its index, and write

$$V \subset B_r(x), \quad h = (h^1, \dots, h^d) : B_{2r}(x) \longrightarrow \mathbb{R}^d.$$

Let h_i be the corresponding harmonic recoveries and let u_i^a be spectral recoveries of u^a . Set

$$Z_i^a = J_i Du_i^a, \quad Q_{a\alpha}^i = \langle Z_i^a, Dh_i^\alpha \rangle, \quad G_{\alpha\beta}^i = \langle Dh_i^\alpha, Dh_i^\beta \rangle. \quad (5.22)$$

On $B_{3r/2}(x_i)$, the local form of (3.9) gives

$$|DQ_{a\alpha}^i| \leq \operatorname{Lip}(h_i^\alpha) |\operatorname{Hess} u_i^a| + \operatorname{Lip}(u_i^a) |\operatorname{Hess} h_i^\alpha|.$$

The bounds in theorem 5.3 and the spectral recovery bounds therefore make $Q_{a\alpha}^i$ locally bounded in $W^{1,2}$. Local Rellich compactness gives strong L^2 -precompactness on $B_r(x_i)$. All convergences here are local on $B_r \Subset B_{3r/2}$, or equivalently are tested after inserting compatible cutoffs equal to one on B_r . Weak tensor convergence of J_i , together with strong convergence of Du_i^a and Dh_i^α , identifies every weak limit as

$$\langle ADu^a, Dh^\alpha \rangle = 0.$$

Thus the whole sequence satisfies

$$Q_i \longrightarrow 0 \quad \text{strongly in } L^2(B_r). \quad (5.23)$$

Fix $0 < \eta < 1/2$ and define

$$\Omega_{i,\eta} = \{ \lambda_{\min}(G_i) \geq \eta, \lambda_{\max}(G_i) \leq \eta^{-1} \}. \quad (5.24)$$

On this set let P_i be the orthogonal projection onto $\operatorname{span}\{Dh_i^1, \dots, Dh_i^d\}$. The Gram matrix of the horizontal parts of the Z_i^a 's is

$$H_i := (\langle P_i Z_i^a, P_i Z_i^b \rangle)_{ab} = Q_i G_i^{-1} Q_i^T. \quad (5.25)$$

Consequently

$$\|H_i\|_{L^1(\Omega_{i,\eta})} \leq \eta^{-1} \|Q_i\|_{L^2(B_r)}^2 \longrightarrow 0.$$

Put $V_i^a = Z_i^a - P_i Z_i^a$ on $\Omega_{i,\eta}$, and denote the Gram matrices of the Z_i^a 's and V_i^a 's by R_i and S_i , respectively. The V_i^a 's lie in the orthogonal complement of a d -dimensional subspace of TM_i , which has dimension $2n - d = c$. Hence $\det S_i = 0$. Since $R_i = S_i + H_i$, all entries are uniformly bounded, and the determinant is Lipschitz on bounded subsets of Sym_{c+1} , we obtain

$$\|\det R_i\|_{L^1(\Omega_{i,\eta})} \longrightarrow 0.$$

Orthogonality of J_i gives

$$R_i^{ab} = \langle Z_i^a, Z_i^b \rangle = \langle Du_i^a, Du_i^b \rangle. \quad (5.26)$$

We finally pass from the moving sets $\Omega_{i,\eta}$ to the limit. Choose a continuous function $\chi_\eta : \text{Sym}_d^{\geq 0} \rightarrow [0, 1]$ which is one when

$$\lambda_{\min} \geq 2\eta, \quad \lambda_{\max} \leq (2\eta)^{-1},$$

and zero when $\lambda_{\min} \leq \eta$ or $\lambda_{\max} \geq \eta^{-1}$. Then $\chi_\eta(G_i)$ is supported in $\Omega_{i,\eta}$, so

$$\chi_\eta(G_i) \det R_i \longrightarrow 0 \quad \text{strongly in } L^1(B_r).$$

By [theorem 5.3](#), $\chi_\eta(G_i) \rightarrow \chi_\eta(G)$ strongly in every finite L^p . The spectral gradients converge strongly in L^2 and are uniformly bounded; hence the entries in (5.26), and therefore their determinants, converge strongly in every finite L^p . It follows that

$$\chi_\eta(G) \det(\langle Du^a, Du^b \rangle)_{a,b=1}^{c+1} = 0 \quad \text{m-a.e. on } B_r.$$

The matrix G is positive definite almost everywhere on V . Letting $\eta \downarrow 0$ proves that the last determinant vanishes almost everywhere on V . The sets V_ℓ cover the d -regular set up to a null set, and that regular set has full measure. Thus $Du^1, \dots, Du^{c+1}$ are linearly dependent almost everywhere on X . Applying this conclusion to the finitely many $(c+1)$ -subsets of a basis of the finite-dimensional space K_0 proves $r_g \leq c$. $\square$

Choose a point x in the positive-measure set on which the integer-valued ghost rank equals r_g , and also in every conull good set used below. Write

$$\text{rank } A_x = 2a, \quad z = \dim \ker A_x = d - 2a. \quad (5.27)$$

Then $r_g \leq \min(c, z)$. Applying [theorem 5.2](#) to an orthonormal basis of K_0 , using [theorem 5.1](#), gives the first principal dimension bound:

$$\boxed{\dim K_0 \leq 2r_g \leq 2 \min(c, z).} \quad (5.28)$$

6 The visible Lie algebra

Let H be the connected analytic subgroup generated by $\mathfrak{v}$ inside $\text{Iso}_{\mathfrak{m}}(X)$, and let

$$K = \overline{H}. \quad (6.1)$$

The group K is compact and connected. The subgroup H need not be closed; only its closure is used as a compact transformation group. Every element of H preserves $\mathfrak{m}$. If $h_j \in H$ converges uniformly to $k \in K$, then for every $\varphi \in C(X)$,

$$\int_X \varphi \circ k \, d\mathbf{m} = \lim_{j \rightarrow \infty} \int_X \varphi \circ h_j \, d\mathbf{m} = \int_X \varphi \, d\mathbf{m}.$$

Thus K also acts by measure-preserving isometries, and the principal orbit theorem [9] gives a conull principal set. Density and $\text{Ad}_H \mathfrak{v} = \mathfrak{v}$ give

$$\mathfrak{v} \triangleleft \mathfrak{k} := \text{Lie } K. \quad (6.2)$$

Choose x as at the end of the preceding section, and additionally so that it is d -regular, K -principal, belongs to the common speed set of [theorem 4.2](#), and is a Lebesgue point for every finite Gram and bracket matrix used below. Put

$$\begin{aligned} e_x : \mathfrak{v} &\longrightarrow T_x X, & e_x(u) &= W_u(x), \\ \mathfrak{h} &= \ker e_x = \mathfrak{v} \cap \mathfrak{k}_x, & D &= e_x(\mathfrak{v}), & k &= \dim D. \end{aligned} \quad (6.3)$$

The equality $\ker e_x = \mathfrak{v} \cap \mathfrak{k}_x$ follows from (4.12)–(4.13).

Lemma 6.1 (Visible dimension at a principal point). *If $\text{rank } A_x = 2a$, then*

$$\boxed{\dim \mathfrak{v} \leq a^2 + 2a.} \quad (6.4)$$

Proof. For $g \in K_x$, define

$$R_g(e_x u) = e_x(\text{Ad}_g u).$$

This is well-defined because conjugation carries a flow fixing x to a flow fixing x . The pointwise norm on D , not merely the L^2 -norm on potentials, is K_x -invariant:

$$\begin{aligned} |e_x(\text{Ad}_g u)| &= N_x(\text{Ad}_g u) \\ &= \lim_{s \rightarrow 0} \frac{d(g \exp(su) g^{-1} x, x)}{|s|} = N_x(u) = |e_x u|. \end{aligned} \quad (6.5)$$

Differentiating gives an orthogonal isotropy representation

$$\rho : \mathfrak{h} \longrightarrow \mathfrak{so}(D), \quad \rho(z)e_x u = e_x[z, u]. \quad (6.6)$$

This representation is faithful. Choose an Ad_K -invariant inner product on $\mathfrak{k}$ and let

$$\mathfrak{v} = \mathfrak{h} \oplus \mathfrak{m}, \quad \mathfrak{m} = \mathfrak{v} \cap \mathfrak{h}^\perp.$$

Then $e_x : \mathfrak{m} \rightarrow D$ is an isomorphism. Since $\mathfrak{h}$ is a Lie subalgebra and the inner product is invariant,

$$\langle [z, w], z' \rangle = -\langle w, [z, z'] \rangle = 0 \quad (z, z' \in \mathfrak{h}, w \in \mathfrak{m}),$$

and therefore $[\mathfrak{h}, \mathfrak{m}] \subset \mathfrak{m}$.

Let $\mathfrak{n} = \ker \rho$. If $z \in \mathfrak{n}$ and $w \in \mathfrak{m}$, then

$$e_x[z, w] = \rho(z)e_x w = 0.$$

Thus $[z, w] \in \mathfrak{h}$; by the preceding inclusion it also belongs to $\mathfrak{m}$, so $[z, w] = 0$. Moreover, ρ is a Lie-algebra homomorphism, and hence

$$\rho([z', z]) = [\rho(z'), \rho(z)] = 0 \quad (z' \in \mathfrak{h}, z \in \mathfrak{n}).$$

It follows that $[\mathfrak{h}, \mathfrak{n}] \subset \mathfrak{n}$, and therefore $\mathfrak{n} \triangleleft \mathfrak{v}$.

Consequently $\text{Ad}_H \mathfrak{n} = \mathfrak{n}$, and continuity of the adjoint action together with the density of H in K gives $\text{Ad}_K \mathfrak{n} = \mathfrak{n}$. The chosen point x is principal. For every other principal point y , there is $g \in K$ such that $K_y = gK_xg^{-1}$; thus

$$\mathfrak{n} = \text{Ad}_g \mathfrak{n} \subseteq \text{Ad}_g \mathfrak{k}_x = \mathfrak{k}_y.$$

The principal set is conull. Hence, for every $z \in \mathfrak{n}$, the isometry $\exp(tz)$ fixes a conull, and therefore dense, subset of X for every t . It is the identity isometry, so its infinitesimal field is $W_z = 0$. Thus $z \in K_0 \cap \mathfrak{v} = \{0\}$, proving $\mathfrak{n} = 0$.

For $u, v \in \mathfrak{v}$, define a two-form on D by

$$\sigma_x(e_x u, e_x v) = \{u, v\}(x). \quad (6.7)$$

It is well-defined. Indeed, if $z \in \mathfrak{h}$, then $\Phi_t^z x = x$, whereas $t \mapsto U_t^z v \in E$ is C^1 in the finite-dimensional Lipschitz eigenspace. Therefore

$$\{z, v\}(x) = \left. \frac{d}{dt} \right|_{t=0} (U_t^z v)(x) = 0. \quad (6.8)$$

Jacobi and (6.8) give

$$\sigma_x(\rho(z)p, q) + \sigma_x(p, \rho(z)q) = 0. \quad (6.9)$$

Let $R = \text{Ran } A_x$, so $\dim R = 2a$, and let $A_R = A_x|_R$. The skew map A_R is invertible and $D \subset R$. For $p = A_x Du$ and $q = A_x Dv$, the R -component of Dv is $P_R Dv = A_R^{-1} q$. Therefore

$$\begin{aligned} \sigma_x(p, q) &= \langle A_x Du, Dv \rangle = \langle p, P_R Dv \rangle \\ &= \langle p, A_R^{-1} q \rangle = \langle -A_R^{-1} p, q \rangle. \end{aligned} \quad (6.10)$$

Thus the skew-adjoint operator representing $\sigma_x|_D$ is

$$C = P_D(-A_R^{-1})|_D \in \mathfrak{so}(D). \quad (6.11)$$

Equations (6.6) and (6.9) imply

$$\rho(\mathfrak{h}) \subset Z_{\mathfrak{so}(D)}(C). \quad (6.12)$$

Write

$$\text{rank } C = 2b, \quad t = \dim \ker C, \quad k = 2b + t. \quad (6.13)$$

The form

$$\Omega(p, q) = \langle -A_R^{-1} p, q \rangle$$

is symplectic on R , and

$$\ker C = D \cap D^\Omega.$$

Since $\dim D^\Omega = 2a - (2b + t)$,

$$t \leq 2a - (2b + t), \quad b + t \leq a. \quad (6.14)$$

If the distinct positive singular values of C have complex multiplicities $m_1, \dots, m_s$, with $\sum_j m_j = b$, its orthogonal normal form gives

$$Z_{\mathfrak{so}(D)}(C) \cong \mathfrak{so}(t) \oplus \bigoplus_j \mathfrak{u}(m_j).$$

By faithfulness of ρ ,

$$\dim \mathfrak{h} \leq \frac{t(t-1)}{2} + \sum_j m_j^2 \leq \frac{t(t-1)}{2} + b^2. \quad (6.15)$$

Consequently,

$$\begin{aligned} \dim \mathfrak{v} &= k + \dim \mathfrak{h} \\ &\leq b^2 + 2b + \frac{t(t+1)}{2} \\ &\leq (b+t)^2 + 2(b+t) \leq a^2 + 2a, \end{aligned}$$

which proves (6.4). $\square$

Corollary 6.2 (Full-rank refinement). *Suppose $\dim \operatorname{Ran} A_x = 2n$. If $k = \dim D < 2n$, then*

$$\dim \mathfrak{v} \leq n^2. \quad (6.16)$$

Proof. Retain b, t from (6.13). If $t = 0$, then $b \leq n - 1$, and

$$\dim \mathfrak{v} \leq b^2 + 2b \leq n^2 - 1.$$

If $t \geq 1$, set $s = b + t \leq n$. The right side obtained from (6.15) is

$$\begin{aligned} F(s, t) &= (s-t)^2 + 2(s-t) + \frac{t(t+1)}{2} \\ &= s^2 + 2s - 2st + \frac{3}{2}t^2 - \frac{3}{2}t. \end{aligned}$$

For fixed t , this is increasing for $s \geq t$, and therefore

$$F(s, t) - n^2 \leq F(n, t) - n^2 = (t-1) \left(\frac{3}{2}t - 2n \right) \leq 0,$$

because $1 \leq t \leq n$. $\square$

7 Collapse is impossible

At the selected point x , combine (5.28) and (6.4):

$$\begin{aligned} \dim E &= \dim K_0 + \dim \mathfrak{v} \\ &\leq 2 \min(c, z) + a^2 + 2a \\ &\leq c + z + a^2 + 2a. \end{aligned} \quad (7.1)$$

By (2.7) and (5.27),

$$c + z = (2n - d) + (d - 2a) = 2(n - a),$$

so

$$\dim E \leq a^2 + 2n. \quad (7.2)$$

If $c > 0$, then $c + z = 2(n - a)$ is a positive even integer. Hence $a \leq n - 1$, and

$$\dim E \leq (n - 1)^2 + 2n = n^2 + 1 < n^2 + 3.$$

This contradicts (2.6). We have proved

$$\boxed{d = 2n.} \quad (7.3)$$

8 Identification of the noncollapsed limit

8.1 Full visible orbit rank

The essential dimension in (7.3) agrees with Honda's Ricci-limit dimension. Since it equals $2n$, the dimension of every approximating manifold, the sequence is noncollapsed in Honda's sense. Thus [12, Theorem 6.19] applies to the weak tensor limit A :

$$J_i \longrightarrow A \quad \text{strongly in } L^2, \quad \nabla A = 0, \quad A^* A = I, \quad A^2 = -I. \quad (8.1)$$

In particular, $\text{rank } A = 2n$ almost everywhere.

Since $c = 0$, [theorem 5.4](#) gives $r_g = 0$, and (5.28) gives

$$K_0 = 0, \quad \dim \mathfrak{v} = \dim E \geq n^2 + 3. \quad (8.2)$$

Choose a principal common-speed point at which $\text{rank } A = 2n$. If $\dim e_x(\mathfrak{v}) < 2n$, [theorem 6.2](#) would imply $\dim \mathfrak{v} \leq n^2$, contradicting (8.2). Therefore

$$\dim e_x(\mathfrak{v}) = 2n. \quad (8.3)$$

8.2 A full-dimensional orbit has positive measure

We give the metric argument explicitly, because a full-dimensional orbit is not automatically open in a nonsmooth space.

Lemma 8.1 (Positive-measure orbit). *Let Z be a compact metric space, V a finite-dimensional connected Lie group, and $\iota : V \rightarrow \text{Iso}(Z)$ a continuous homomorphism. Suppose:*

- (a) *the orbit map is locally Lipschitz in the V -variable;*
- (b) *for every $\xi \in \mathfrak{v}$,*

$$\lim_{s \rightarrow 0} \frac{d(\iota(\exp(s\xi))x, x)}{|s|} = |e_x \xi|;$$

- (c) $\text{rank } e_x = d_0$.

Then

$$\mathcal{H}^{d_0}(\iota(V)x) > 0. \quad (8.4)$$

Proof. Let

$$V_x = \{g \in V : \iota(g)x = x\}.$$

It is a closed Lie subgroup. By the zero-speed implication (4.18),

$$\text{Lie}(V_x) = \ker e_x. \quad (8.5)$$

Hence $\dim(V/V_x) = d_0$.

Choose an injective smooth quotient chart $\psi : Q \rightarrow V/V_x$, where $Q \subset \mathbb{R}^{d_0}$ is a small cube, and a smooth local lift $s : Q \rightarrow V$. Define

$$F : Q \longrightarrow Z, \quad F(t) = \iota(s(t))x. \quad (8.6)$$

The map is Lipschitz by (a), and injective because ψ is injective. Let

$$\eta_t(v) = (dL_{s(t)^{-1}})_{s(t)}(ds_t v) \in \mathfrak{v}. \quad (8.7)$$

The induced map $\bar{\eta}_t : \mathbb{R}^{d_0} \rightarrow \mathfrak{v}/\text{Lie}(V_x)$ is an isomorphism. Smooth Lie-group calculus gives

$$s(t)^{-1}s(t+rv) = \exp(r\eta_t(v))q_r, \quad d_V(q_r, e) = o(|r|). \quad (8.8)$$

Local Lipschitzness of the orbit map turns the q_r -term into an $o(|r|)$ displacement. Isometric invariance and the speed formula therefore give

$$\lim_{r \rightarrow 0} \frac{d(F(t+rv), F(t))}{|r|} = |e_x \eta_t(v)|. \quad (8.9)$$

Kirchheim's metric differentiability theorem [13] applies to F . At almost every t , compare its metric differential first on a countable dense set of directions with (8.9), and then use continuity in v , to obtain

$$\text{md } F_t(v) = |e_x \eta_t(v)| \quad (v \in \mathbb{R}^{d_0}). \quad (8.10)$$

This is a norm. Indeed, if its value at v is zero, then $\eta_t(v) \in \ker e_x = \text{Lie}(V_x)$; since $\bar{\eta}_t$ is an isomorphism, $v = 0$.

Shrink the quotient chart so that Q has compact closure in its original domain. The function

$$(t, v) \longmapsto |e_x \eta_t(v)|$$

is continuous and positive on $\bar{Q} \times \mathbb{S}^{d_0-1}$. Its minimum is therefore a positive number c_0 , and for almost every $t \in Q$,

$$\text{md } F_t(v) \geq c_0 |v| \quad (8.11)$$

for all $v \in \mathbb{R}^{d_0}$. Kirchheim's area formula and injectivity of F now yield

$$\mathcal{H}^{d_0}(F(Q)) = \int_Q J_{d_0}(\text{md } F_t) \, dt \geq c \mathcal{L}^{d_0}(Q) > 0.$$

□

We verify the local Lipschitz hypothesis for H . Choose a basis $u_1, \dots, u_r$ of $\mathfrak{v}$. Finite dimensionality and the uniform eigenfunction gradient estimate give $\|W_u\|_\infty \leq C\|u\|$. Hence (4.16) gives

$$d(\Phi_t^u y, y) \leq C|t| \|u\| \quad (y \in X).$$

In canonical product-of-exponentials coordinates, a telescoping argument, using that each factor acts isometrically, makes every orbit map locally Lipschitz with a constant uniform in the base point. Applying theorem 8.1 to H , with $d_0 = 2n$ and (8.3), gives

$$\mathcal{H}^{2n}(Hx) > 0, \quad \mathcal{H}^{2n}(Kx) > 0. \quad (8.12)$$

8.3 Homogeneity and the exact rigidity theorem

Since the essential dimension equals the synthetic upper dimension $2n$, [3, Theorem 2.20] gives weak noncollapse, and [3, Theorem 1.3] upgrades it to

$$\mathfrak{m} = C \mathcal{H}^{2n} \quad (8.13)$$

for some $C > 0$. Thus (8.12) implies $\mathfrak{m}(Kx) > 0$. As recorded after (6.1), K is a closed group of measure-preserving isometries. Proposition 5.10 of [14] now gives

$$Kx = X. \quad (8.14)$$

By [14, Proposition 5.13], the homogeneous RCD metric space (X, d) is isometric to a smooth homogeneous Riemannian manifold. The K -invariant probability measure on K/K_x is the push-forward of normalized Haar measure and is unique. Since both $\mathfrak{m}$ and normalized Riemannian volume are K -invariant, they are proportional.

On this smooth manifold, $A \in L^\infty(T_1^1 X)$ satisfies $\nabla A = 0$ distributionally by (8.1). In smooth local coordinates this is a linear first-order system with smooth coefficients, so A agrees almost everywhere with a smooth parallel tensor. The identities $A^*A = I$ and $A^2 = -I$ then hold everywhere. Thus A is a parallel orthogonal almost-complex structure; its Nijenhuis tensor vanishes and its fundamental two-form is parallel, so (X, g, A) is Kähler. Because $\mathfrak{m}$ is proportional to Riemannian volume, the smooth interpretation of $\text{RCD}(1, 2n)$ gives $\text{Ric}_g \geq g$. Finally, (2.6) says that L_X has at least $n^2 + 3$ real 2-eigenfunctions. In the complex normalization,

$$\lambda_{n^2+3}(-\Delta_{\omega_A}) = 1.$$

The exact rigidity theorem [7, Theorem 1.6] therefore yields

$$(X, g, A) \cong (\mathbb{CP}^n, \omega_{\mathbb{CP}^n}, J_{\text{std}}) \quad \text{biholomorphically and isometrically.} \quad (8.15)$$

9 Return to the approximating manifolds

The normalized measures in (2.5) do not by themselves determine the absolute volumes. However, that convergence includes the underlying Gromov–Hausdorff convergence, and we now know that its limit is smooth and has the same real dimension $2n$ as the approximating manifolds. Colding’s volume convergence theorem [8], applied at this point, therefore gives convergence of the unnormalized Riemannian volumes:

$$\text{vol}(M_i, \omega_i) \longrightarrow \text{vol}(\mathbb{CP}^n, \omega_{\mathbb{CP}^n}). \quad (9.1)$$

Set

$$\eta_i = \frac{\omega_i}{n+1}, \quad \omega_{\text{FS}} = \frac{\omega_{\mathbb{CP}^n}}{n+1}. \quad (9.2)$$

The Ricci form is unchanged by constant scaling, so

$$\text{Ric}(\eta_i) \geq (n+1)\eta_i, \quad \text{vol}(M_i, \eta_i) \longrightarrow \text{vol}(\mathbb{CP}^n, \omega_{\text{FS}}). \quad (9.3)$$

Zhang–Liu use $\text{Vol}(M, \omega) = \int_M \omega^n = n! \text{vol}_g(M)$; hence (9.3) gives the same convergence in their convention. For all sufficiently large i , the volume-gap theorem [15, Theorem 1.3] implies

$$M_i \simeq \mathbb{CP}^n \quad \text{biholomorphically.} \quad (9.4)$$

At the same time, (2.5) and (8.15) give

$$d_{\text{GH}}((M_i, \omega_i), (\mathbb{CP}^n, \omega_{\mathbb{CP}^n})) \longrightarrow 0. \quad (9.5)$$

Equations (9.4)–(9.5) contradict (2.3). This proves [theorem 1.1](#). $\square$

Dependency audit

For clarity, the proof does *not* rely on any of the following statements:

- strong convergence of J_i in the collapsed regime;
- preservation of the tensor A by visible flows in the collapsed regime;
- closedness of the analytic subgroup H ;
- the assertion that a full-dimensional orbit is open;
- convergence of unnormalized volumes from normalized pmGH convergence alone.

The replacements are, respectively: strong scalar bracket compactness; the intrinsic two-form σ_x ; passage to $K = \overline{H}$; the quotient-chart/area-formula argument; and Colding's volume convergence after noncollapse and smoothness of the limit are known.

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